

John McGahay

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Curriculum Vitae

Education

- 2027
(expected) **Doctor of Philosophy in Linguistics**
University of California, Los Angeles | Los Angeles, CA, USA
- Dissertation:* Confusion Minimization Theory: Towards a unified computational account of sound change and phonological typology based on optimal listeners and speakers
- Committee:* Bruce Hayes (co-chair), Megha Sundara (co-chair), David Goldstein, Connor Mayer, Laurel Perkins
- 2024 **Master of Arts in Linguistics**
University of California, Los Angeles | Los Angeles, CA, USA
- Thesis:* Vowel system typology and confusion minimization
- Committee:* Bruce Hayes (co-chair), Megha Sundara (co-chair), Meg Cychosz
- 2019 **Bachelor of Arts, summa cum laude**
University of Pennsylvania | Philadelphia, PA, USA
- Majors:* Linguistics (with honors)
Cognitive Science (Language and Mind concentration)
- Minors:* Computer Science
German
- 2015 **High School Diploma**
Regis High School | New York, NY, USA

Publications & Manuscripts

Peer-reviewed publications

CRAM, CORALIE; JOHN MCGAHAY; and MEGHA SUNDARA. 2026. Vowel allophony improves maximum-likelihood classification of Warlpiri consonants. *Proceedings of Interspeech 2026*: 7090–7095.

MCGAHAY, JOHN. 2025. Modeling vowel system typology using Iterated Confusion Minimization. *Proceedings of Interspeech 2025*: 2955–2959.

Manuscripts under review

McGAHAY, JOHN. 2026. A theory of vowel system typology based on confusion minimization. Manuscript under review (revised and resubmitted).

McGAHAY, JOHN. 2026. Optimization of English vowels for word-level perception predicts dialect-specific susceptibility to the *cot-caught* merger. Manuscript submitted to ICPhS 2027.

Manuscripts in preparation

McGAHAY, JOHN. Reconceptualizing phonetic contrast and effort: An information-theoretic approach. Manuscript in preparation.

McGAHAY, JOHN. A computational model of allophony as dispersion over phonetic sequences. Manuscript in preparation.

HARRIS, JESSE; CHRISTIAN MUXICA; and **JOHN McGAHAY.** The neighborhood has gone to h*ll: Competition from high frequency lexical neighbors in pupillometry. Manuscript in preparation.

Academic theses

McGAHAY, JOHN. 2024. Vowel system typology and confusion minimization. Los Angeles, CA, USA: University of California, Los Angeles MA thesis.

McGAHAY, JOHN. 2019. The effect of conditioned mergers on underlying representations. Philadelphia, PA, USA: University of Pennsylvania BA honors thesis.

Presentations

Conference talks

McGAHAY, JOHN. 2027. Optimizing US English vowels for word-level perception predicts the *cot-caught* merger. Talk to be presented at the annual meeting of the Linguistic Society of America, San Francisco, CA, USA.

McGAHAY, JOHN. 2026. A computational model of allophony as dispersion over phonetic sequences. Talk presented at the annual meeting of the Linguistic Society of America, New Orleans, LA, USA.

McGAHAY, JOHN. 2025. Modeling vowel system typology using iterated confusion minimization. Talk presented at Interspeech 2025, Rotterdam, The Netherlands.

McGAHAY, JOHN. 2019. The effect of conditioned mergers on underlying representations. Talk presented at the 43rd Penn Linguistics Conference, Philadelphia, PA, USA.

Invited talks

McGAHAY, JOHN. 2026. A computational theory of vowel system typology based on optimal listeners and speakers. Invited talk presented to the UC Berkeley PhonLab, Berkeley, CA, USA.

Conference posters

CRAM, CORALIE; **JOHN McGAHAY**; and MEGHA SUNDARA. 2026. Vowel allophony improves maximum-likelihood classification of Warlpiri consonants. Poster presented at Interspeech 2026, Sydney, NSW, Australia.

CRAM, CORALIE; **JOHN McGAHAY**; and MEGHA SUNDARA. 2026. Vowel inventory size moderates the contribution of vowel allophony to consonant classification: Evidence from Warlpiri and Dalabon. Poster presented at Interspeech 2026, Sydney, NSW, Australia.

McGAHAY, JOHN. 2026. A computational model of merger as phonetic dispersion over words. Poster presented at LabPhon 20, Montréal, QC, Canada.

McGAHAY, JOHN. 2026. A Bayesian account of perceptual asymmetries in confusion matrix data. Poster presented at the 190th meeting of the Acoustical Society of America, Philadelphia, PA, USA.

BOULOM, ATLAS; CORALIE CRAM; NICHOLAS GUYMON; **JOHN McGAHAY**; ZACHARY METZLER; EMMA MONTILLA; VISHWAS SHETTY; JIAN-LEAT SIAH; and MEG CYCHOSZ. 2025. A systematic review and meta-analysis of the development of coarticulation in child speech. Poster presented at the 188th Meeting of the Acoustical Society of America, New Orleans, LA, USA.

McGAHAY, JOHN. 2023. A theory of vowel dispersion based on probabilistic modeling of optimized speakers and listeners. Poster presented at the 185th Meeting of the Acoustical Society of America, Sydney, NSW, Australia.

Grants & Awards

AY 2026-2027	Dissertation Year Award (\$33,500) <i>UCLA Graduate Education</i>
AY 2025-2026	Graduate Research Mentorship (\$30,000) <i>UCLA Graduate Education</i>
AY 2025-2026	Pauley Fellowship (\$17,000) <i>UCLA Division of Humanities</i>
Summer 2025	Graduate Summer Research Mentorship (\$6,000) <i>UCLA Graduate Education</i>

AY 2024-2025	Graduate Research Mentorship (\$20,000) <i>UCLA Graduate Education</i>
Summer 2024	Summer Mentored Research Award (\$6,000) <i>UCLA Graduate Education</i>
Spring 2024	Biggs Graduate Fellowship (\$10,500) <i>UCLA Department of Linguistics</i>
Summer 2023	Graduate Summer Research Mentorship (\$6,000) <i>UCLA Graduate Education</i>
AY 2022-2023	Pauley Fellowship (\$17,000) <i>UCLA Division of Humanities</i>
AY 2019-2020	Fullbright Award (€8,273) <i>Fulbright Germany</i>
March 2019	Best Undergraduate Abstract (\$250) <i>Penn Linguistics Conference 43</i>
Summer 2018	Jumpstart for Juniors (\$1,000) <i>Penn Center for Undergraduate Research & Fellowships</i>
Summer 2016	Penn Undergraduate Research Mentorship (\$3,500) <i>Penn Center for Undergraduate Research & Fellowships</i>
April 2016	Otto Springer Prize (\$250) <i>Penn Department of Germanic Languages and Literatures</i>

Teaching Experience

Teaching assistantships at the University of California, Los Angeles

Fall 2024	LING 185A: Computational Linguistics I <i>Instructor: Prof. Laurel Perkins</i>
Spring 2024	LING C111: Intonation <i>Instructor: Prof. Sun-Ah Jun</i>
Winter 2024	LING 165A: Phonology II <i>Instructor: Prof. Sam Zukoff</i>
Winter 2024	LING 102: Introduction to Applied Phonetics <i>Instructor: Dr. Elise Bell</i>

Fall 2023 **LING 103: Introduction to General Phonetics**
Instructor: Prof. Sun-Ah Jun

Teaching assistantship at the University of Pennsylvania

Fall 2018 **LING 336: Sound Systems in Use** (corpus sociophonetics)
Instructor: Prof. Meredith Tamminga

Other teaching experience

Sept 2019 – **Fulbright English Teaching Assistant**
March 2020 *Nicolaus-Cusanus-Gymnasium | Bonn, NRW, Germany*

Sept 2015 – **German Tutor**
May 2019 *Penn Language Center | Philadelphia, PA, USA*

Research Assistant Experience

Summer 2026 **Research Assistant to Prof. Jonah Katz**
UCLA Linguistics Department | Los Angeles, CA, USA

- Praat scripting for acoustic analysis of lenition patterns in English and Spanish

December 2024 **Research Assistant to Prof. Meg Cychosz and Dr. Alejandrina Cristia**
Laboratoire de Sciences Cognitives et Psycholinguistique | Paris, France

- Processing LENA recordings of infants' linguistic input using NLP pipelines

Summer 2024 **Research Assistant to Prof. Jesse Harris**
UCLA Language Processing Lab | Los Angeles, CA, USA

- Phonetic manipulation of stimuli for pupillometry experiment

Summer 2023 **Research Assistant to Prof. Sun-Ah Jun**
UCLA Phonetics Lab | Los Angeles, CA, USA

- ToBI intonation transcription of experiment stimuli and subject responses

May 2016 – **Research Assistant to Prof. Meredith Tamminga**
May 2019 *Penn Language Variation and Cognition Lab | Philadelphia, PA, USA*

- Maintenance and reorganization of the Philadelphia Neighborhood Corpus
- Processing/analysis of phonetic data using Praat scripting, R, Python, ELAN

February 2016 – **Research Assistant to Prof. Jianjing Kuang**
May 2016 *Penn Phonetics Lab | Philadelphia, PA, USA*

- Segmentation and labeling of Shanghainese wordlist recordings in Praat

Other Work Experience

- June 2021 – July 2022 **Junior Computational Linguist**
Verbio Technologies | Remote
- Developing intent-classification systems used for call center automation
 - Bash scripting for manipulation of large training datasets
- March 2021 – June 2021 **Natural Language Understanding Specialist**
Cerence | Remote
- Training and testing of language models used for car voice control features
- April 2020 – August 2021 **Computational Linguistics Temporary Worker**
Linguistic Data Consortium | Remote
- Assistant on project to develop a new forced aligner
 - Transcription of clinical speech for neurodegenerative disease research
- May 2019 – August 2019 **Computational Linguistics Temporary Worker**
Linguistic Data Consortium | Philadelphia, PA, USA
- Phonetic/phonological analysis of Portuguese dialects for ASR applications
 - Implementation of grammars in OpenFst-based packages Pynini and Thrax

Selected Skills

- Technical:* Python, Praat (incl. scripting), R, Bash, LaTeX, Microsoft Office, Stan, Labvanced, Haskell, OCaml, Java, MATLAB, ELAN
- Language:* English (native); German (fluent); French, Spanish (intermediate); Irish, Portuguese, Russian, Italian, American Sign Language (basic)

References

Bruce Hayes, Ph.D.
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Megha Sundara, Ph.D.
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